Central Training & Placement Cell, AKTU in
association with ERA Foundation presents

KALAM PRAGATI

Empowering Engineers with Skills for Success

MakeX Internship

RAPID PROTOTYPE & DESIGN THIMKING
REPORT - 2025

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Project Overview

Project Title	CardioGuard – Real-Time Heart Monitoring and Alert System	
Team Name	Solution Squad	
Team Members	Name	Roll Number
	Ayushman Srivastav	2301870480003
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	Shivansh Srivastava	2300560100203
Problem identified	Sudden heart attacks and cardiac emergencies often go undetected until it is too late. Lack of real-time monitoring and timely alerts leads to delayed medical response, increasing the risk of complications and death.	
Solution proposed	CardioGuard, a wearable ECG-based monitoring system integrated with a multi-dashboard alert network (patient, family, doctor, ambulance, and hospital). It provides real-time ECG monitoring, automatic alerts, location tracking, and pre-hospital notifications to save golden minutes in emergencies.	
Link to the idea pitch presentation	link	
Link to photos drive	link	

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1. Project Overview

1.1 Abstract

CardioGuard is a real-time heart monitoring and alert system developed to address the increasing risk of sudden cardiac deaths caused by delayed recognition and treatment of heart attacks. The solution integrates a **wearable ECG strip** with an automated **multi-dashboard ecosystem** that connects patients, families, doctors, ambulances, and hospitals. By enabling continuous monitoring, instant SOS alerts, and location tracking, the system helps save critical golden minutes during emergencies.

Throughout the internship, our team applied design thinking methodologies such as **5W1H, WHY Analysis, Problem Tree & Ideation (Crazy 8)** to systematically analyze the problem and design an impactful solution. Field visits with doctors, pharmacists, and patients validated the need for such a system and guided the prototype design.

The project not only demonstrated the feasibility of **IoT-enabled health monitoring and AI-based predictive alerts** but also enhanced our skills in problem analysis, prototyping, and teamwork, making this a highly valuable learning experience.

1.2 Introduction

Heart disease remains one of the leading causes of sudden deaths worldwide, particularly in India. Despite advancements in healthcare, many patients fail to receive timely medical attention during cardiac emergencies. Often, symptoms are mistaken for minor issues, leading to fatal delays in treatment. Our project ‘CardioGuard’ was developed to address this challenge by creating a real-time monitoring and alert system that can automatically detect abnormal heart activity and notify families, doctors, and ambulances immediately.

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1.3 Proposed Solution

CardioGuard, a wearable ECG-based monitoring system integrated with a multi-dashboard alert network (patient, family, doctor, ambulance, and hospital). It provides real-time ECG monitoring, automatic alerts, location tracking, and pre-hospital notifications to save golden minutes in emergencies.

1.4 Mind map

■ Heart Saver (CardioGuard) – Mind Map

Problem ■ • Heart hurts suddenly • No quick help	**Who Needs Help? ■■■■■■■** • Patients • Elders • Families • Doctors
What To Do? ■ • Wear strip • Alarm beep • Call family & doctor • Send location	**Goal ■** • Save lives • Quick help • Less worry

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2. Problem Statement

2.1 Problem Statement

How might we help patients, their families, doctors, and ambulance drivers move from delayed, manual, and uncertain responses to real-time, automated, and connected alerts with live data and location, so as to save golden minutes, reduce sudden deaths, and ensure timely treatment?

2.2 What is the Problem?

Heart attacks occur suddenly and unpredictably. Patients often do not receive early warnings or immediate medical help. Existing systems like smartwatches or hospital ECG machines either lack accuracy, portability, or real-time emergency alerts.

2.3 Why is it a Problem?

Delayed recognition and treatment of heart attacks lead to severe complications and death. Families and patients feel helpless, doctors face late arrivals, and ambulances are often called too late.

2.4 Who is Facing the Problem?

- Heart patients (especially elderly and high-risk individuals)
- Families and caregivers
- Doctors and hospitals
- Ambulance services

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2.5 When and Where does the Problem Occur?

- At home, especially when the patient is alone
- During travel (bus, train, cab, airplane)
- At workplace or public spaces
- At night, when family is asleep

2.6 Existing Solutions

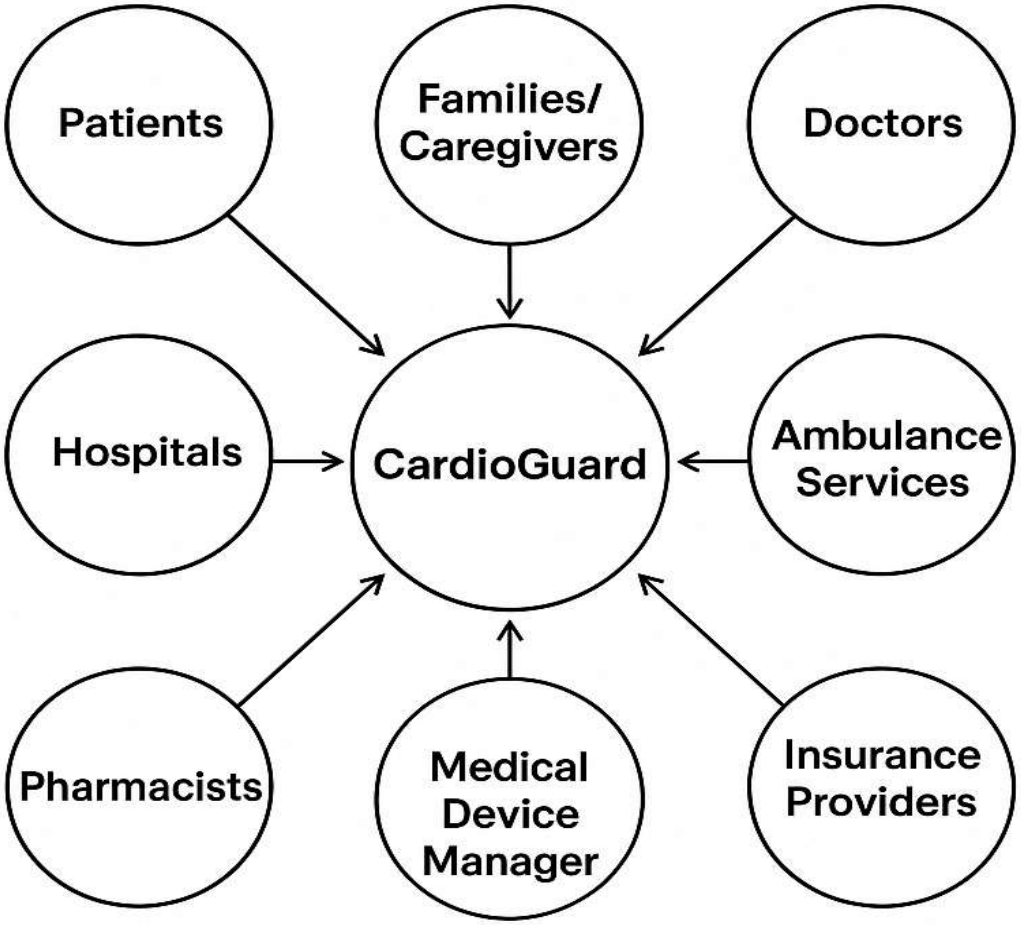
- Smartwatches and fitness bands (inaccurate, limited)
- Hospital ECG machines (accurate but not portable)
- Mobile health apps (lack automatic alerts)
- Ambulance helplines (manual call required)

2.7 Complaints about Existing Solutions

- Accuracy issues and false alarms
- No real-time emergency alerts
- High cost of medical devices
- Limited usability and bulky devices
- Manual effort required in emergencies

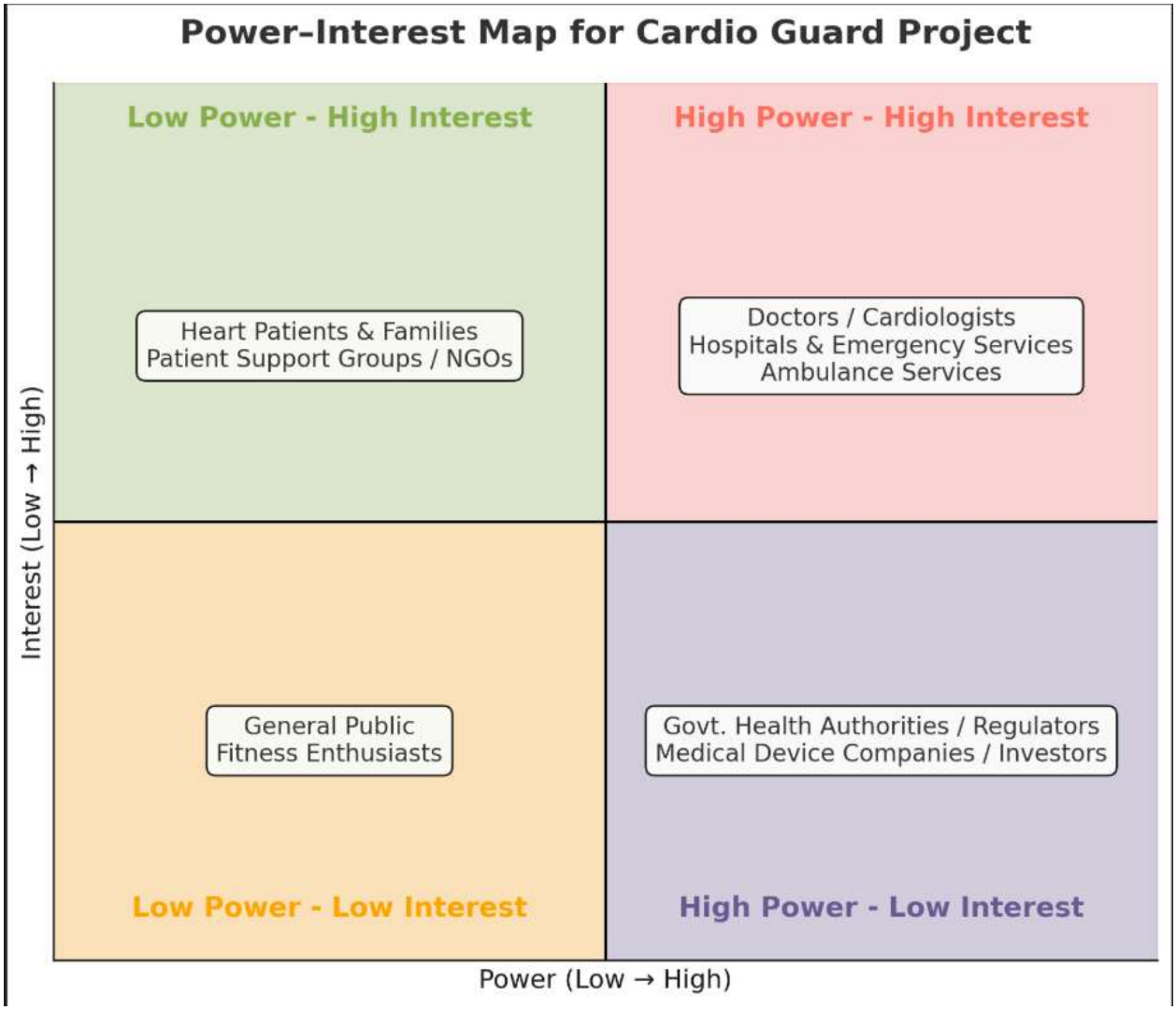
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2.8 Stake Holder Mapping



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2.9 Power Vs Interest map



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3. Field Visits and research

To better understand the problem of delayed recognition and treatment of heart attacks, our team conducted field visits and interacted with multiple stakeholders including pharmacists, doctors, and cardiology experts.

Pharmacists: Reported that many patients purchase heart-related medicines without proper prescriptions, increasing risks due to lack of monitoring.

Doctors: Shared that patients often confuse symptoms of heart attack with gas pain or acidity, leading to dangerous delays in treatment.

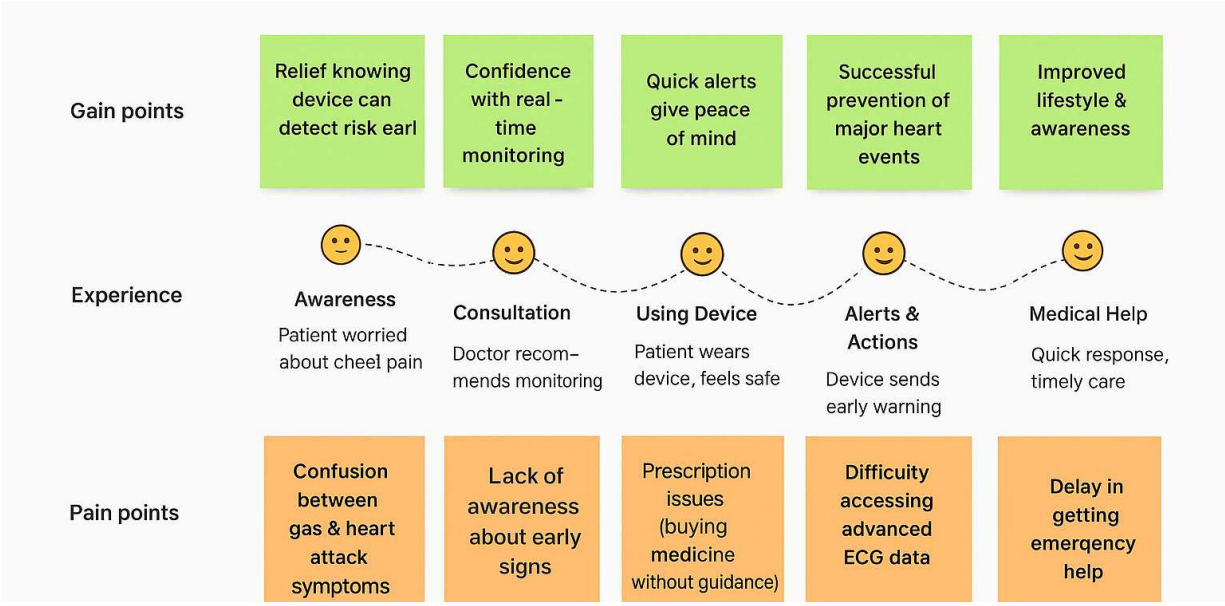
Cardiologists: Highlighted that ST Segment Elevation and Complete Heart Block (CHB) are among the most critical indicators that must be detected in real time for effective early intervention.

These insights validated the need for a real-time monitoring and alert system like CardioGuard

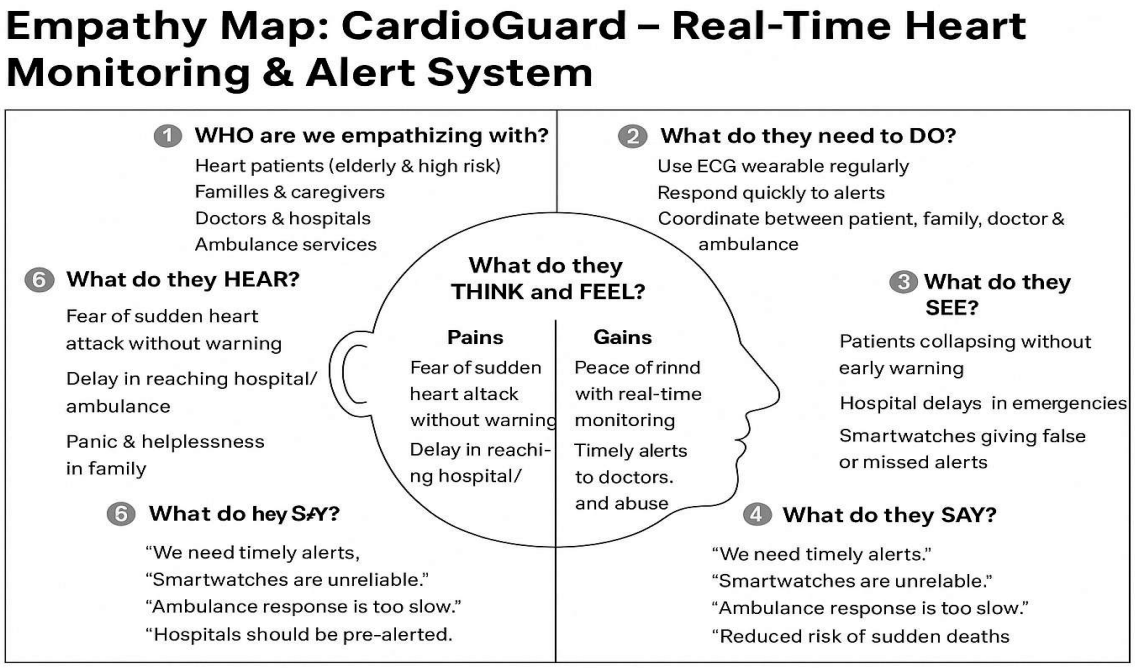


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3.1 Journey Map

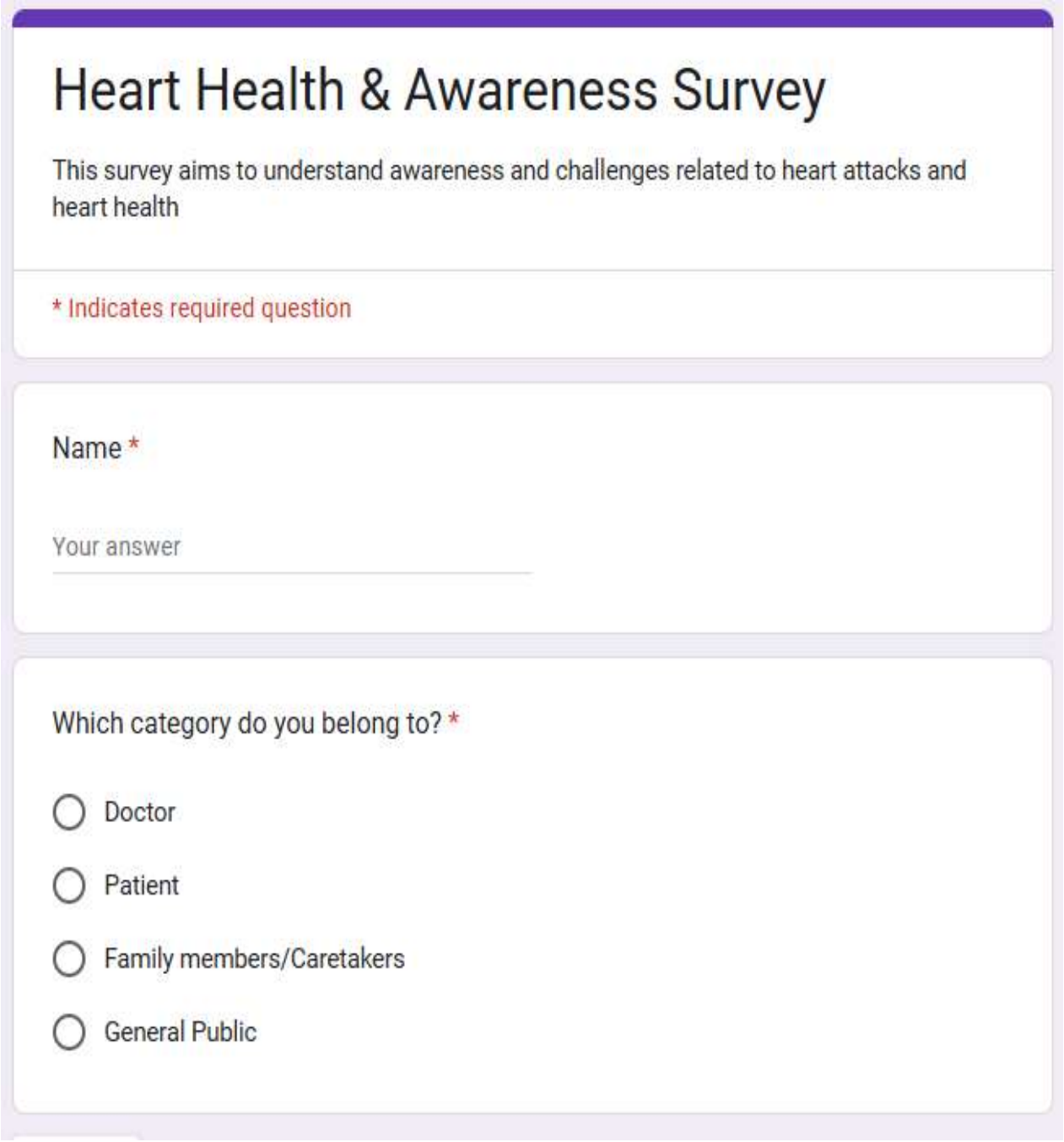


3.2 Empathy Map



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3.3 Survey Forms/ Questionnaires



Heart Health & Awareness Survey

This survey aims to understand awareness and challenges related to heart attacks and heart health

* Indicates required question

Name *

Your answer

Which category do you belong to? *

- Doctor
- Patient
- Family members/Caretakers
- General Public

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Patient

How would you describe your lifestyle ? *

- ◯ Active (daily exercise , walk)
- ◯ Moderate Avila
- ◯ Low activity

How much knowledge do you have about heart attacks symptoms ? *

- ◯ no Knowledge
- ◯ A little
- ◯ Well aware

Do you have a family history of heart disease ? *

- ◯ Yes
- ◯ No
- ◯ Not sure

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Family Members / Caretakers

How do you monitor the patients health ? *

- ◯ Regular checkup
- ◯ No monitor

If the patients has chest pain what would you do first ? *

- ◯ Call the doctor immediately
- ◯ Give medicine at home
- ◯ Do nothing

do you know CPR (cardio pulmonary resuscitation) *

- ◯ Yes
- ◯ No
- ◯ A little

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General Public

Do you think stress affects health ? *

- ○ yes
- ○ no

How often do you exercise? *

- ○ Daily
- ○ Sometimes
- ○ Never

Do you know the symptoms of heart attack *

- ○ yes
- ○ no
- ○ a little

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Doctor

How many cardiac patients do you see daily? *

- ◯ Very high
- ◯ Moderate
- ◯ Very few

Do you think public awareness about heart diseases is sufficient ? *

- ◯ Yes
- ◯ No

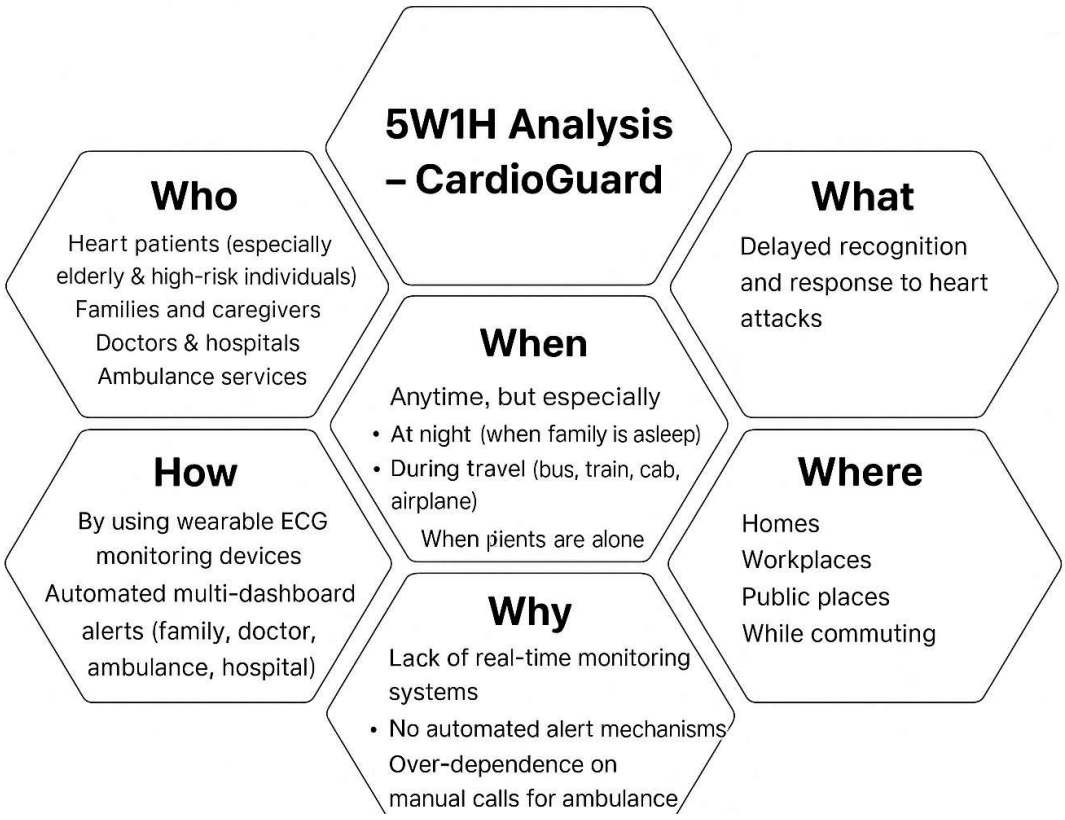
In your opinion what is the biggest risk factor for heart attack ? *

- ◯ Diabetes
- ◯ High BP
- ◯ Smoking
- ◯ Stress

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.4. Problem Analysis

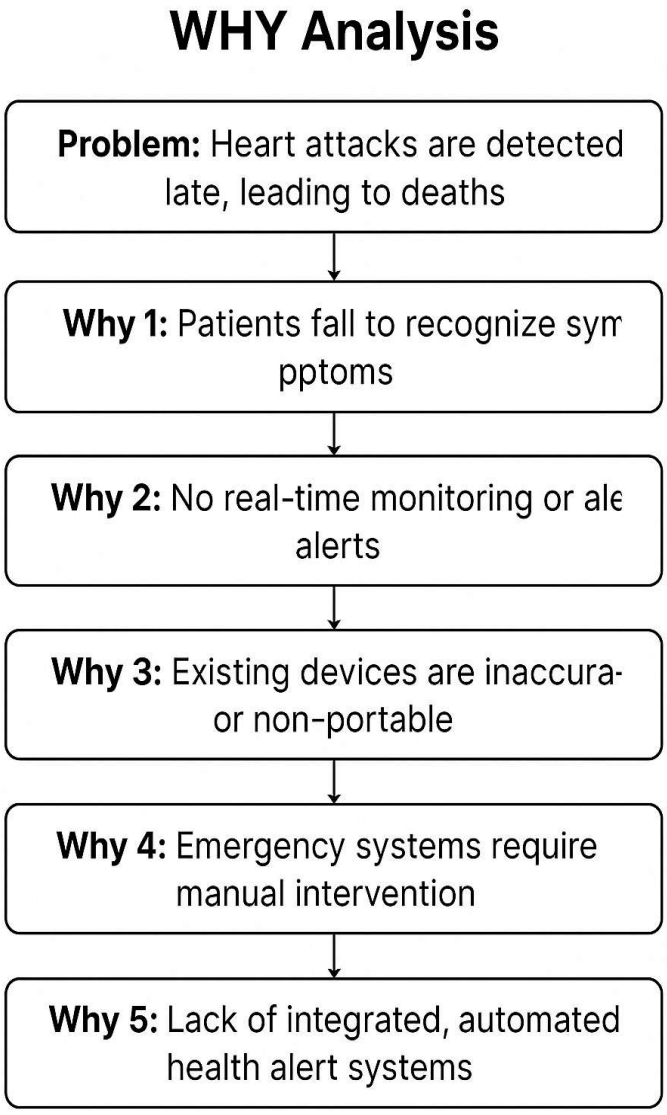
4.1 5W1H Analysis



-**Who:** Heart patients, families, doctors, and ambulance services.
- **What:** Delayed recognition and response to heart attacks.
- **When:** Anytime, but especially at night, during travel, or when patients are alone.
- **Where:** Homes, workplaces, public places, and while commuting.
- **Why:** Lack of real-time monitoring and automatic alert systems.
- **How:** Through wearable ECG monitoring and automated alerts.

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4.2 WHY Analysis



Problem: Heart attacks are detected late, leading to deaths.

Why 1: Patients fail to recognize symptoms.

Why 2: No real-time monitoring or alerts.

Why 3: Existing devices are inaccurate or non-portable.

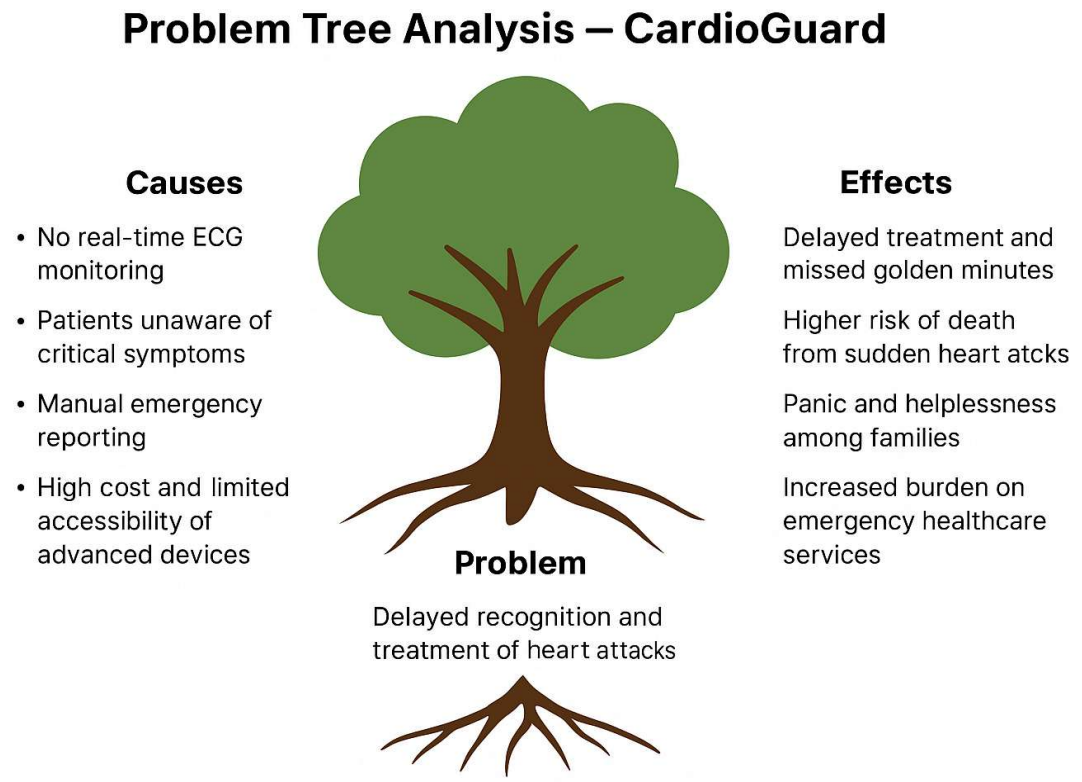
Why 4: Emergency systems require manual intervention.

Why 5: Lack of integrated, automated health alert systems.

Root Cause: Absence of a reliable, real-time heart monitoring and alert solution.

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4.3 Problem Tree Analysis



Causes:

- No real-time ECG monitoring
- Patients unaware of critical symptoms
- Manual emergency reporting
- High cost and limited accessibility of advanced devices

Effects:

- Delayed treatment and missed golden minutes
- Higher risk of death from sudden heart attacks
- Panic and helplessness among families
- Increased burden on emergency healthcare services

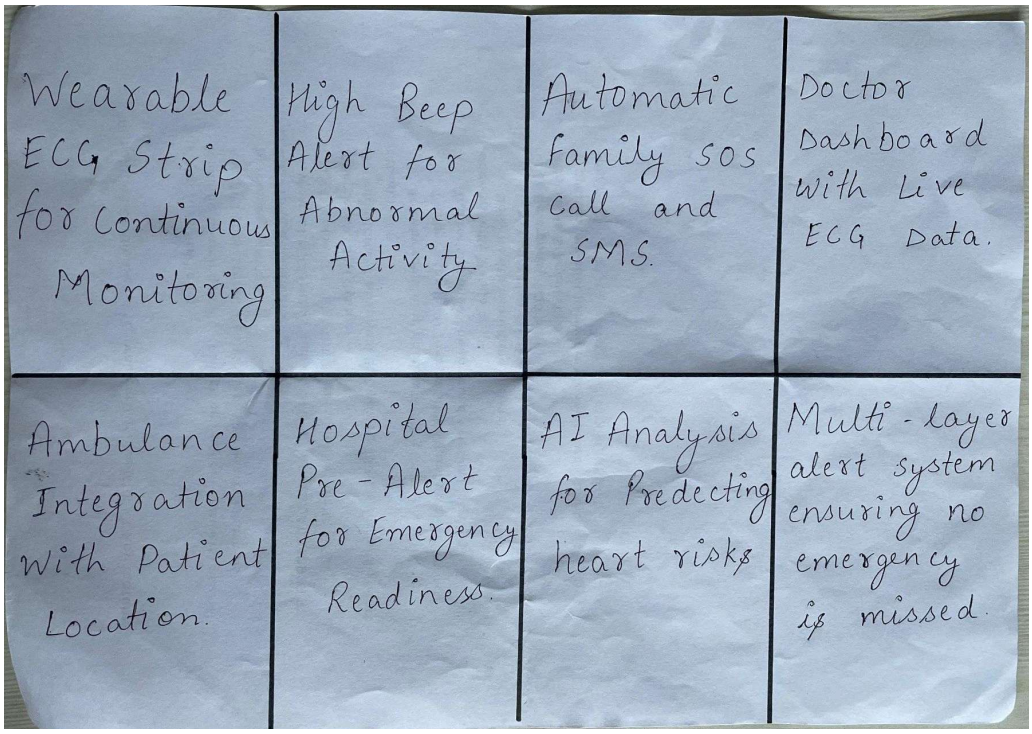
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5. Solution

5.1 Ideation

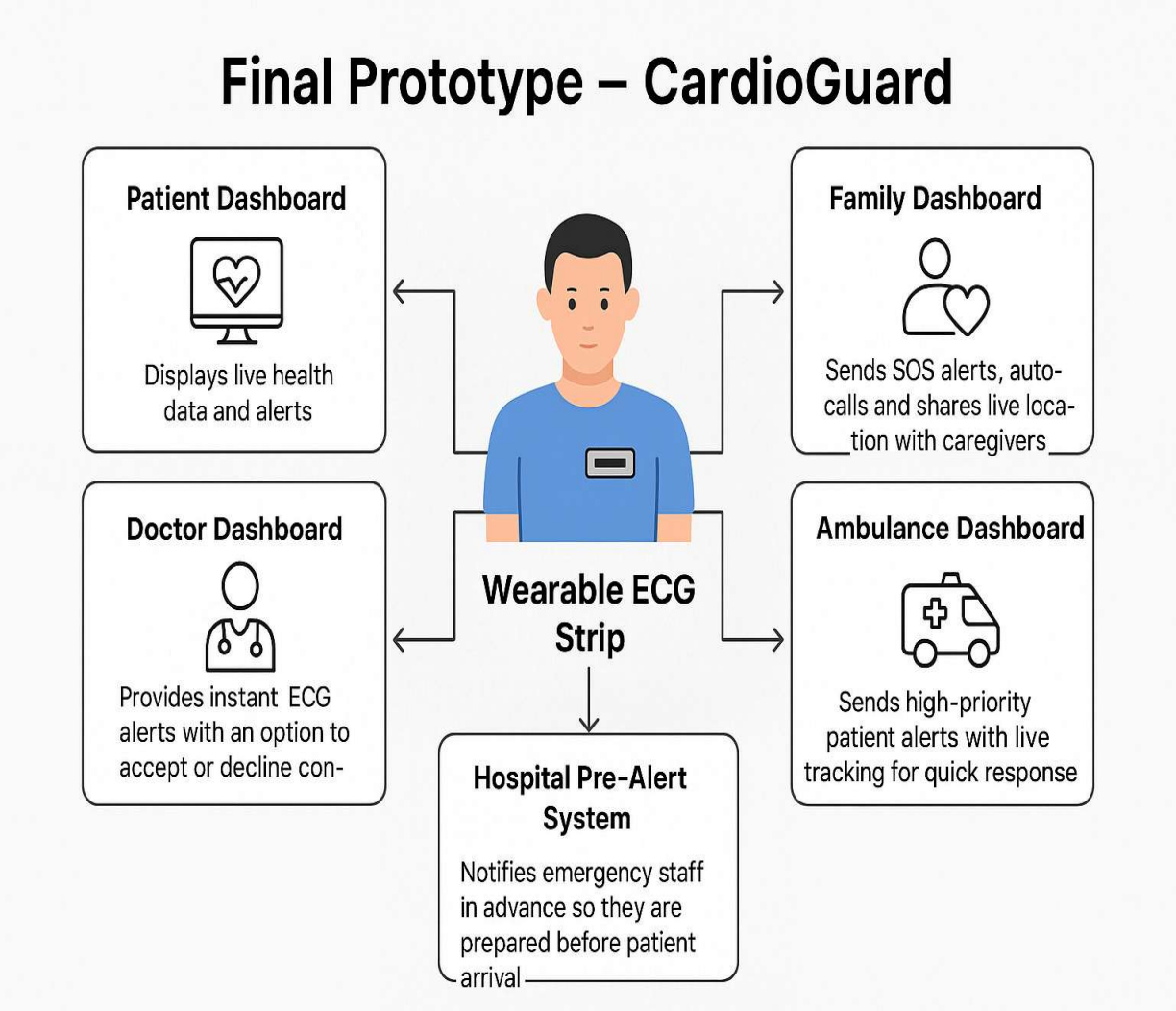
Our team used the ‘Crazy 8’ brainstorming method to generate multiple concepts. Key ideas included:

- Wearable ECG Strip for continuous monitoring
- High Beep Alert for abnormal activity
- Automatic Family SOS Call & SMS
- Doctor Dashboard with live ECG data
- Ambulance Integration with patient location
- Hospital Pre-Alert for emergency readiness
- AI Analysis for predicting heart risks
- Multi-layer alert system ensuring no emergency is missed



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5.2 Proposed Idea

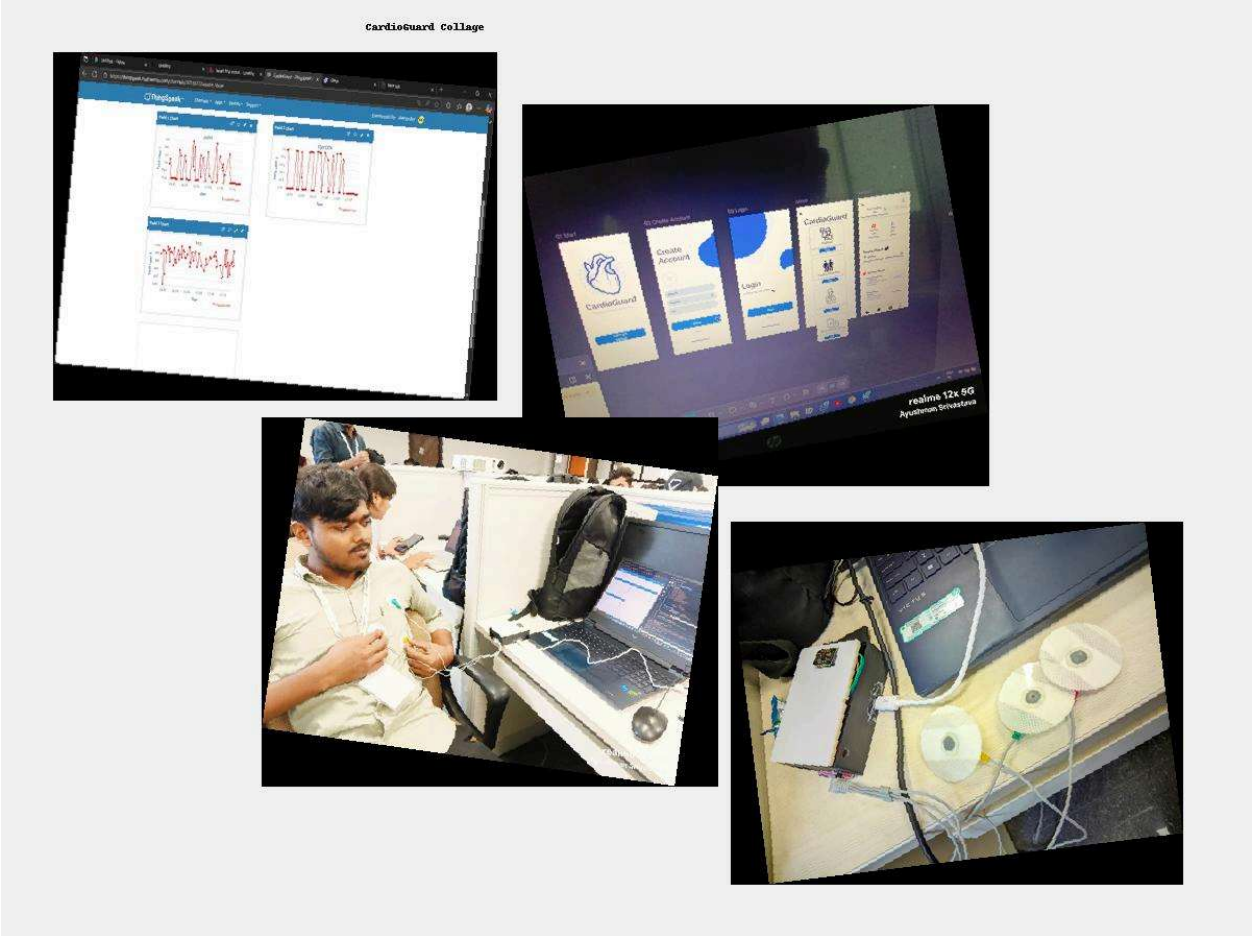


- A wearable ECG strip for real-time monitoring
- Patient Dashboard with live health data
- Family Dashboard for SOS alerts, calls, and location tracking
- Doctor Dashboard with ECG alerts and accept/decline options
- Ambulance Dashboard with high-priority patient alerts
- Hospital Pre-Alert system to prepare ER staff before arrival.

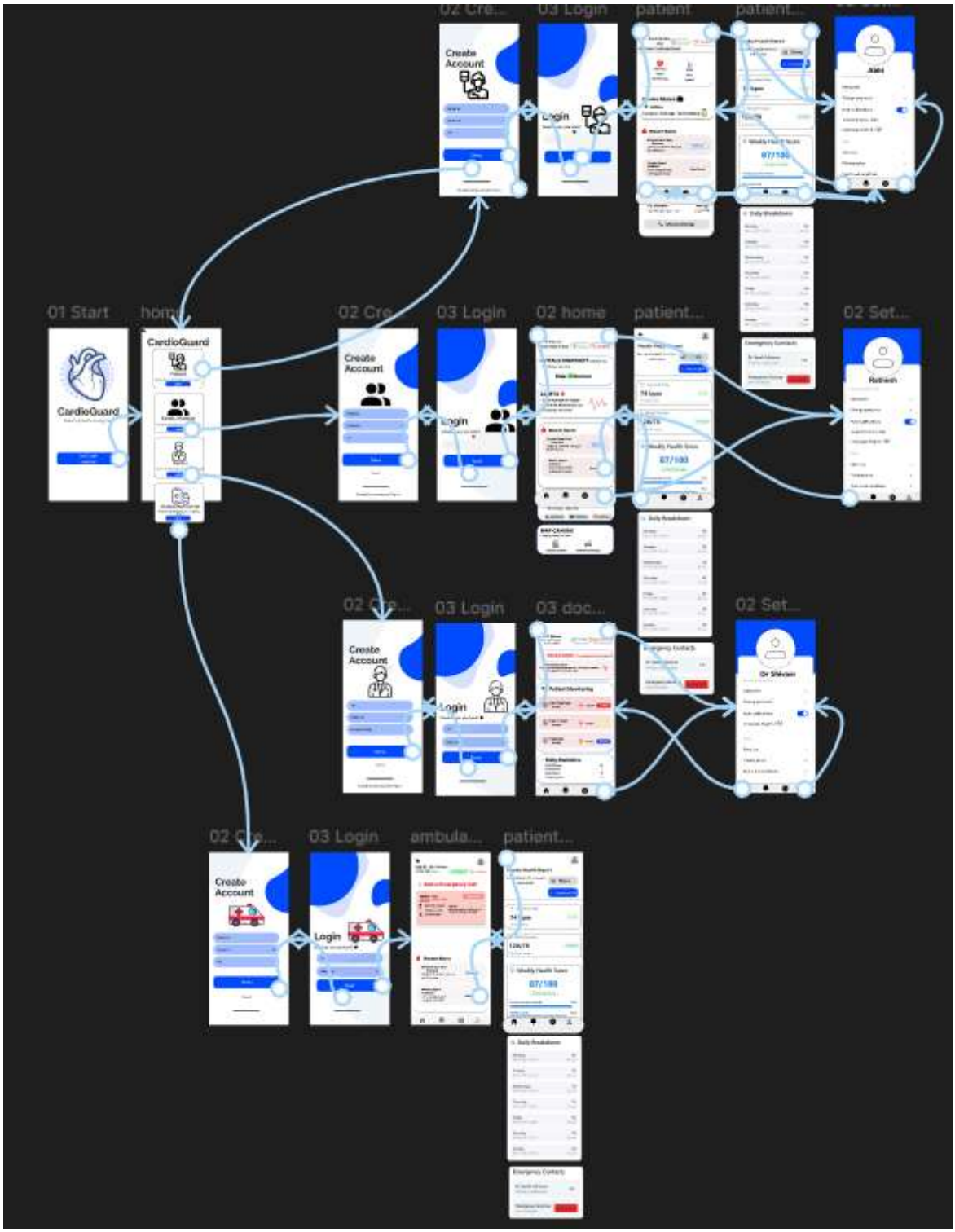
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- Key Partners: Hospitals, ambulance services, cardiologists, device manufacturers
- Key Activities: Continuous monitoring, alert integration, data analysis
- Value Proposition: Saves golden minutes, reduces sudden deaths, ensures timely treatment
- Customer Segments: Patients, families, doctors, hospitals, ambulance services
- Channels: Mobile app, dashboards, wearable device
- Revenue Streams: Device sales, subscription services, hospital partnerships
- Cost Structure: Hardware, software development, cloud hosting, customer support

5.3 Prototype(Hardware/Software/UI Screens)



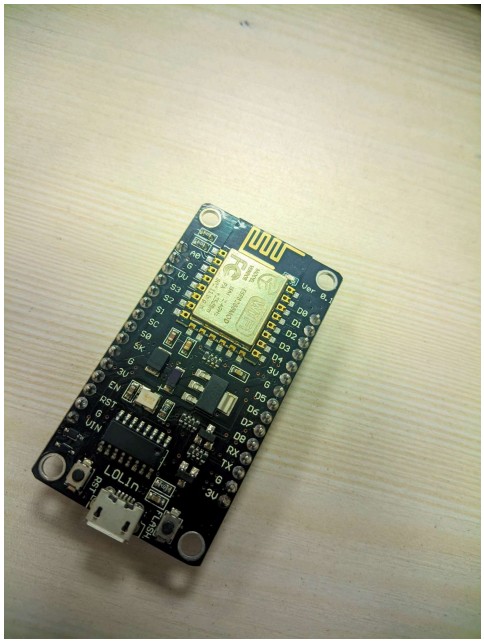
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01 Start
CardioGuard
Create Account
Login
02 home
patient...
02 Set...
03 Login
03 doc...
ambula...

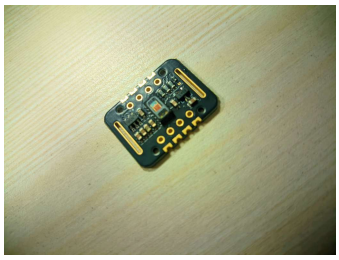
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5.4 Tools & Technoligies Used

Hardware Components

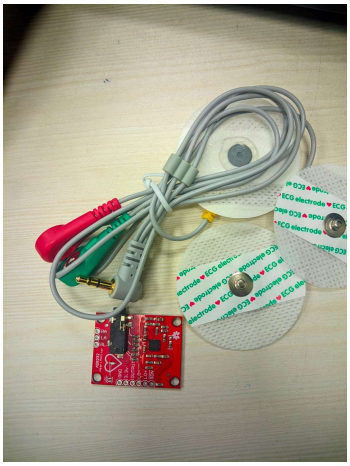


ESP8266 (NodeMCU) – main microcontroller with WiFi for IoT connectivity



MAX30102 Sensor – optical sensor for Heart Rate (BPM) and SpO_2 (Oxygen saturation)

ECG Sensor (AD8232) – for ECG waveform signal



Software / Libraries

Arduino IDE – coding & uploading firmware

ESP8266WiFi.h – WiFi connectivity

ThingSpeak.h – sending sensor data to cloud dashboard

MAX30105.h + heartRate.h – library for MAX30102 sensor

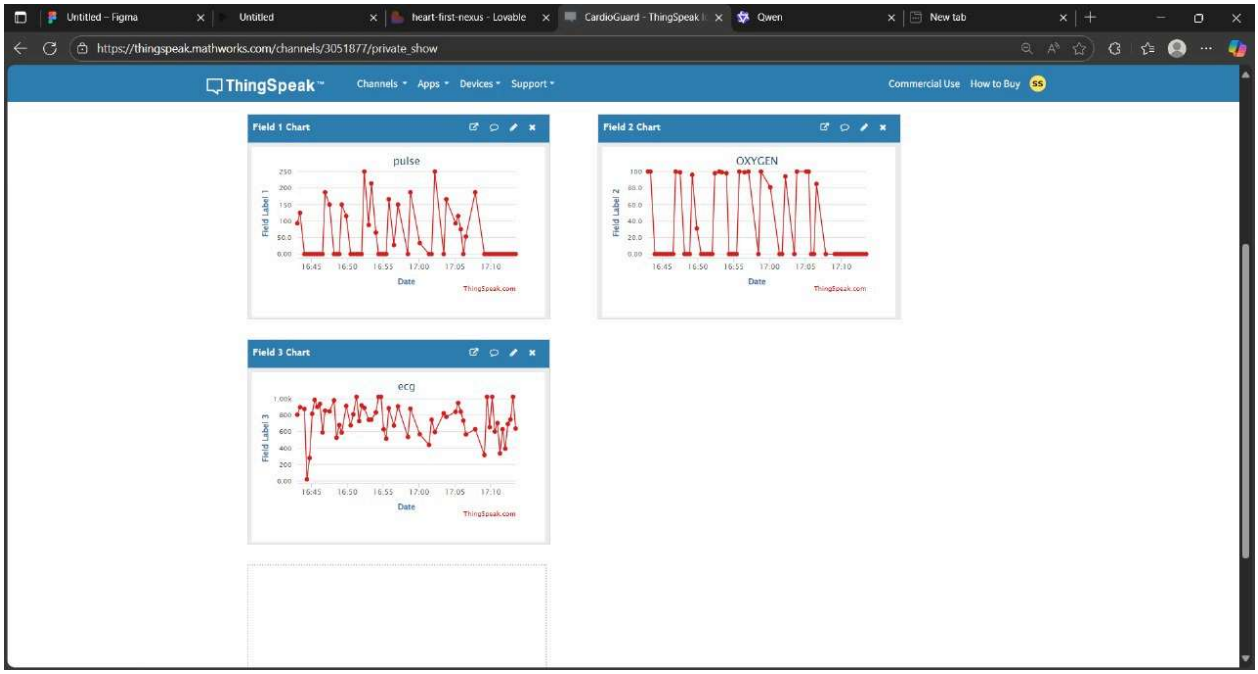
UniversalTelegramBot.h – Telegram bot API communication

WiFiClientSecure.h – SSL/TLS connection for Telegram

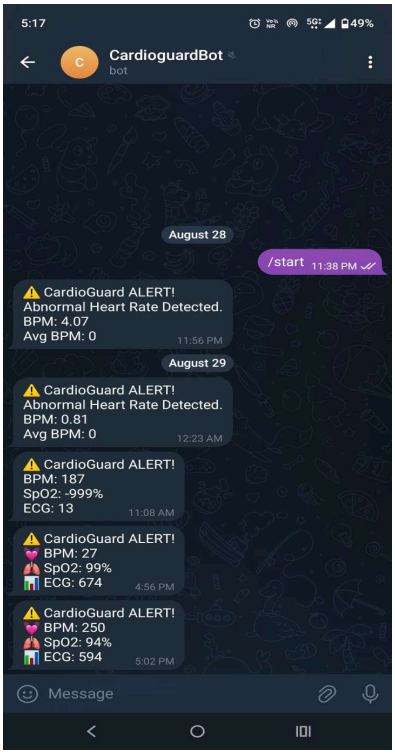
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IoT Platforms

ThingSpeak Cloud – for real-time data logging (BPM, SpO_2, ECG values)



Telegram Bot API – for instant health alerts & notifications

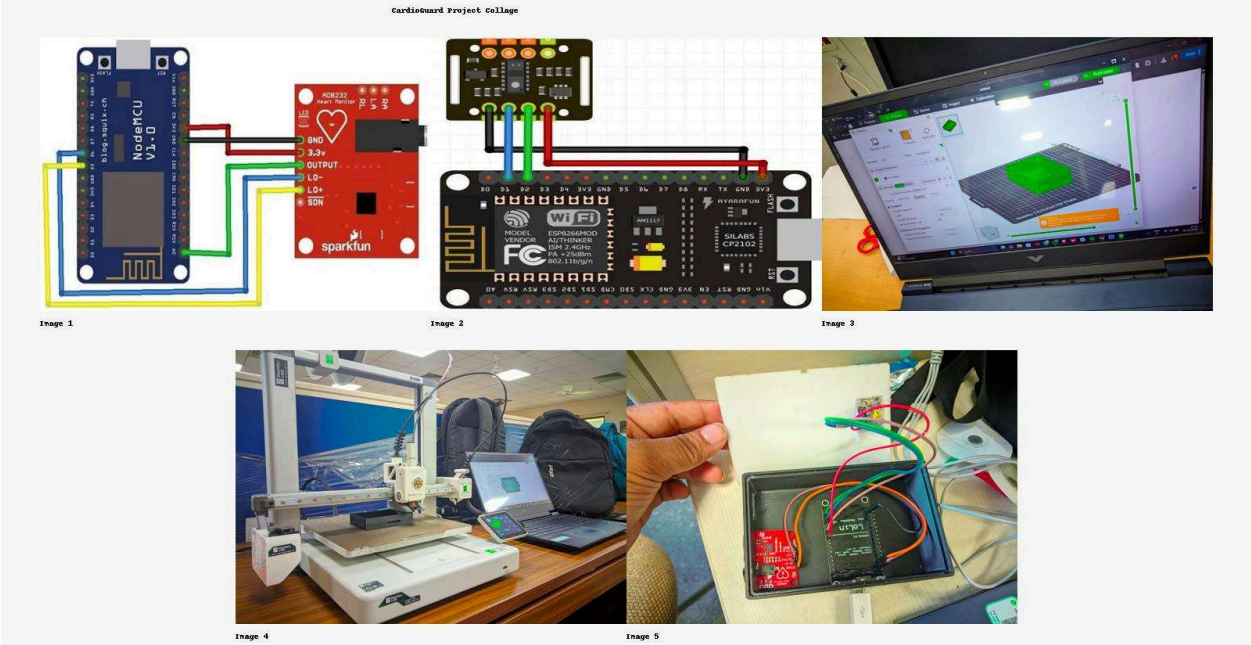


Programming Language

Embedded C / C++ (via Arduino framework)

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5.5 Circuit Diagrams / CAD Designs / Screenshots



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6. Conclusion

CardioGuard provides a transformative solution to the problem of delayed recognition and treatment of heart attacks. By integrating real-time ECG monitoring with automated alerts to families, doctors, ambulances, and hospitals, it saves critical golden minutes and ensures timely treatment. The system not only empowers patients and families but also enhances healthcare efficiency.

6.1 Learning

- Technical: Gained knowledge of ECG signal monitoring, IoT integration, alert systems, and dashboard design.
- Operational: Learned about stakeholder needs and emergency healthcare processes.
- Problem-Solution Fit: Understood how real-time monitoring directly addresses a life-critical challenge.

6.2 Future Scope

- Mobile app with AI-driven predictive alerts
- Integration with insurance providers and healthcare records
- Lightweight wearable versions for higher comfort
- Enhanced AI models for predictive diagnosis
- Large-scale deployment in hospitals and public health systems

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